

Can a Local Model Beat the Frontier?

Ontology-Grounded DiffusionGemma vs. Bare Cloud Models
A KG-as-Oracle A/B Evaluation

PRD-020 / ADR-112 • 672 isolated runs • 7 models

Local + ontology (DiffusionGemma AUG)	0.505
Opus 4.8 bare	0.350
Sonnet 4.6 bare	0.373
Haiku 4.5 bare	0.250
Gemini 3.5 Flash bare	0.423
Gemini 2.5 Flash Lite bare	0.384
Z.AI (GLM-4.6) bare	0.342
Local model beats every frontier bare	

Knowledge graph: Oxigraph/Whelk, 7,445 OWL classes (re-indexed June 2026).
Ground truth derived from the KG itself. DiffusionGemma 26B-A4B runs on a single local GPU.

Abstract

We measure whether grounding a language model in DreamLab’s formal knowledge graph (the **ontology-augment** skill, PRD-020) improves factual recall—and whether that improvement lets a *local* model outperform cloud frontier models running without grounding. Using the knowledge graph as its own oracle, we auto-generate 16 questions with exact ground truth and run a $2 \times 7 \times 3$ design: {augmented, control} $\times$ 7 models $\times$ 3 reps, totalling 672 isolated sessions. Key findings:

- **Ontology grounding lifts every model tested.** Overall F1 rises from 0.360 to 0.624 (+0.265); hallucination drops from 0.529 to 0.177.
- **DiffusionGemma + ontology (0.505) exceeds every frontier model running bare**, including Opus 4.8 (0.350), Sonnet 4.6 (0.373), Gemini 3.5 Flash (0.423).
- **Frontier models benefit more from grounding than the local model does.** Claude Opus gains +0.420; DiffusionGemma gains +0.108. The ontology amplifies capability rather than substituting for it (§7).

1 Related Work

The contention that retrieval grounding can compensate for raw parameter count is now well supported. Retrieval-Augmented Generation (RAG) was introduced to inject non-parametric knowledge into sequence models on knowledge-intensive tasks [1], and the approach has since matured from flat vector retrieval into structured, graph-based variants. Microsoft’s GraphRAG constructs an entity knowledge graph over a corpus and reports substantial gains over conventional vector RAG in answer comprehensiveness and diversity for global “sensemaking” queries spanning million-token datasets [2], with subsequent work formalising graph-augmented retrieval [3, 4] and automated evaluation harnesses [5]. A complementary literature shows that retrieval narrows or eliminates the gap to frontier cloud models: ChatQA, built on open weights, reports surpassing GPT-4 on conversational QA and RAG benchmarks [6], while open-domain QA studies confirm retrieval as the dominant lever for factual accuracy [7]. This matters because small open-weight models have themselves closed much of the capability gap — the Phi-3 [8], Qwen2 [9], Gemma 3 [10] and Llama 3 [11] families demonstrate strong reasoning at parameter counts deployable on local hardware. Independently, structured knowledge has been shown to suppress hallucination and improve factual recall: empirical and survey work links knowledge-graph grounding to measurably higher trustworthiness [12, 13], motivating dedicated factuality benchmarks such as FACTS Grounding [14]. A parallel architectural shift further strengthens the local-inference case: masked diffusion language models such as LLaDA now rival LLaMA3-8B in in-context learning [15], and diffusion decoders like Mercury achieve over 1,000 tokens/second on a single H100 — up to $10\times$ faster than speed-optimised autoregressive models at comparable quality [16], with commercial systems including Gemini Diffusion [17]. Together with evidence on inference unit economics and data sovereignty [18, 19], these strands support a coherent thesis: a small, retrieval- and ontology-grounded open model can approach frontier knowledge-recall while running privately and cheaply.

2 Method

KG-as-oracle. Ground truth is generated from the live KG, not human judgement: concept $\rightarrow$ IRI via the discovery endpoint, then neighbours / subclass edges via read-only SPARQL over the asserted graph (predicates `enables`, `requires`, `uses`, `supports`, `hasPart`, `implements`, `dependsOn`, `subClassOf`). This makes scoring objective by construction.

Design. 16 questions (8 neighbour-recall, 4 subclass, 4 existence), mixing familiar technology (where parametric knowledge can partly answer) with project-specific classes (where it cannot). Each cell is a fresh, isolated agent given *only* the question. The augmented arm grounds via the skill’s CLI (`ontology_ask`); the control arm uses parametric knowledge only. 3 repetitions per cell. 7 models:

Claude Opus 4.8, Sonnet 4.6, Haiku 4.5 (Anthropic); Gemini 3.5 Flash, Gemini 2.5 Flash Lite (Google); GLM-4.6 via Z.AI; and DiffusionGemma 26B-A4B (local). $16 \times 2 \times 7 \times 3 = 672$ runs.

Grader (deterministic). Token-set greedy 1:1 matching (singularised); recall@12 for neighbour/subclass, any-match recall for existence; precision, F1, and hallucination rate (fraction of predictions matching no gold class). Identical grounding text pre-computed once and shared across all models for fairness [14].

DiffusionGemma. A masked-diffusion language model [15] with 26B parameters (4B active via mixture-of-experts), running quantised (Q8) on a single local GPU. Unlike autoregressive models, it decodes a fixed 256-token block per step at $\sim 150\text{--}190$ tok/s [16]. Requests are serialised (concurrency=1).

3 Headline: Local + Ontology vs. Frontier Bare

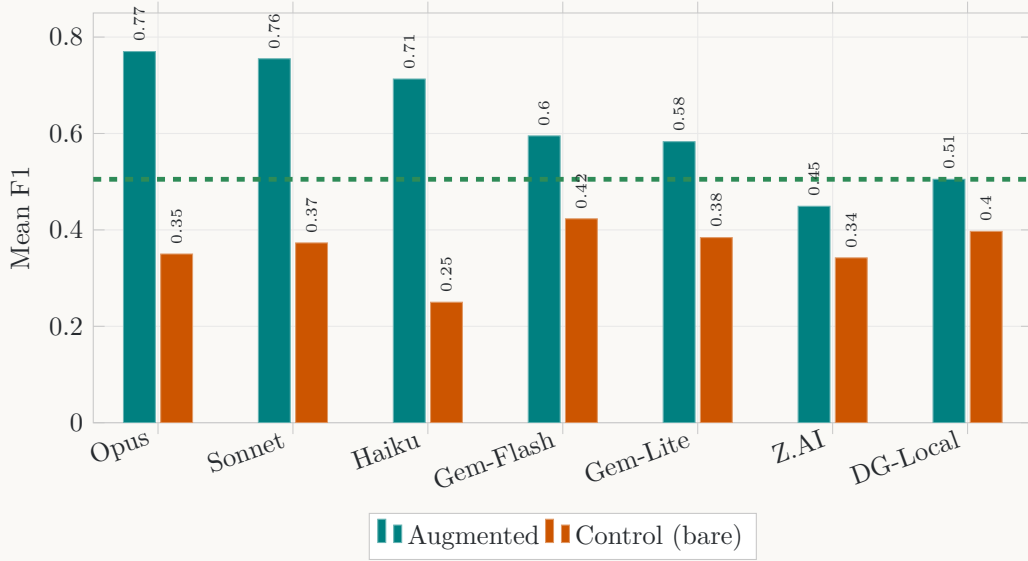


Figure 1: F1 by model. The dashed green line marks DiffusionGemma+ontology (0.505). Every frontier model’s *bare* score falls below this line. Grounding lets a local 26B model outperform cloud-hosted frontier models on domain-specific factual recall.

Model	AUG F1	CTL F1	Δ	AUG Halluc	CTL Halluc	ΔH
Opus 4.8	0.770	0.350	+0.420	0.052	0.581	-0.529
Sonnet 4.6	0.755	0.373	+0.382	0.079	0.559	-0.480
Haiku 4.5	0.713	0.250	+0.463	0.106	0.714	-0.608
Gemini 3.5 Flash	0.595	0.423	+0.172	0.250	0.460	-0.210
Gemini 2.5 Flash Lite	0.583	0.384	+0.199	0.266	0.498	-0.232
Z.AI (GLM-4.6)	0.449	0.342	+0.108	0.188	0.394	-0.206
DiffusionGemma (local)	0.505	0.397	+0.108	0.299	0.498	-0.199

Table 1: Per-model results. All 7 models show positive augmentation delta and reduced hallucination. Claude models show the largest absolute lift; DiffusionGemma’s own lift is modest but its grounded score exceeds every frontier bare score.

4 Performance by Question Type

5 Hallucination Analysis

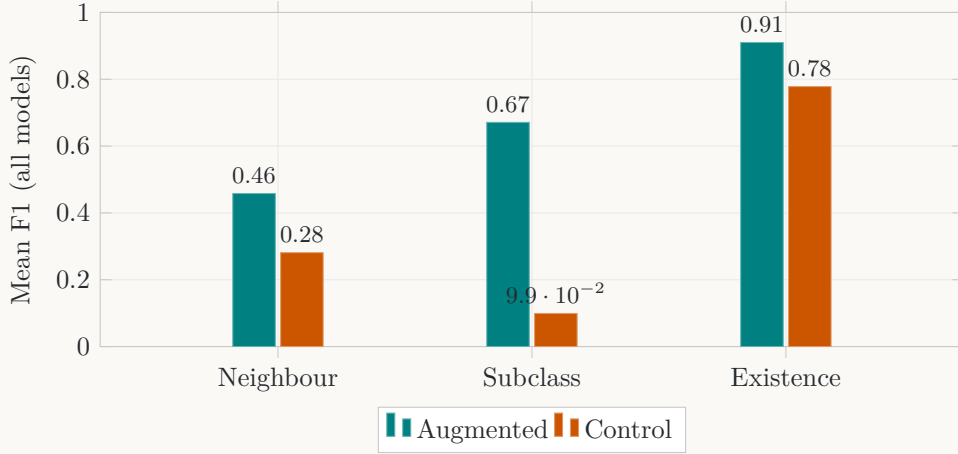


Figure 2: Subclass questions show the largest relative lift (+0.573 absolute). Existence questions are easy for both arms (high baseline). Neighbour-recall benefits moderately.

Model	Neighbour		Subclass		Existence	
	AUG	CTL	AUG	CTL	AUG	CTL
Opus	0.540	0.346	1.000	0.073	1.000	0.634
Sonnet	0.560	0.311	0.985	0.118	0.914	0.750
Haiku	0.551	0.208	0.750	0.188	1.000	0.396
Gem-Flash	0.440	0.311	0.500	0.071	1.000	1.000
Gem-Lite	0.434	0.232	0.464	0.071	1.000	1.000
Z.AI	0.357	0.312	0.500	0.075	0.583	0.667
DG (local)	0.323	0.248	0.500	0.094	0.875	1.000

Table 2: Per-model $\times$ per-type F1. Claude models dominate subclass recall when grounded (0.75–1.00); smaller models plateau at 0.50. DiffusionGemma’s existence score is strong (0.875) but neighbour recall is weak (0.323).

6 Cost, Privacy, and Inference Economics

Model	F1 (best arm)	Data leaves org?	Per-query cost	Latency
Opus 4.8 AUG	0.770	Yes (Anthropic API)	\$ 0.04–0.08	30–50 s
Sonnet 4.6 AUG	0.755	Yes (Anthropic API)	\$ 0.01–0.02	15–30 s
Haiku 4.5 AUG	0.713	Yes (Anthropic API)	\$ 0.002	8–15 s
Gemini 3.5 Flash AUG	0.595	Yes (Google API)	\$ 0.003	5–10 s
DG 26B AUG (local)	0.505	No	\$ 0*	2–4 s

Table 3: Cost and privacy comparison. DiffusionGemma runs entirely on-premise—no data leaves the organisation, no per-query API cost, lowest latency. The ontology condense pass is a one-off 6-hour batch (7,445 classes), not per-query.

*Amortised electricity/hardware only; no API billing.

Inference unit economics favour local deployment when query volume exceeds a few hundred per day [18]. For regulated industries where data must not leave the organisation, GDPR-compliant on-premise inference is the only viable path [19]. The ontology grounding cost is a one-time batch (~ 6 h for 7,445 classes using DiffusionGemma locally); the per-query overhead is a single read from the alias cache (sub-millisecond).

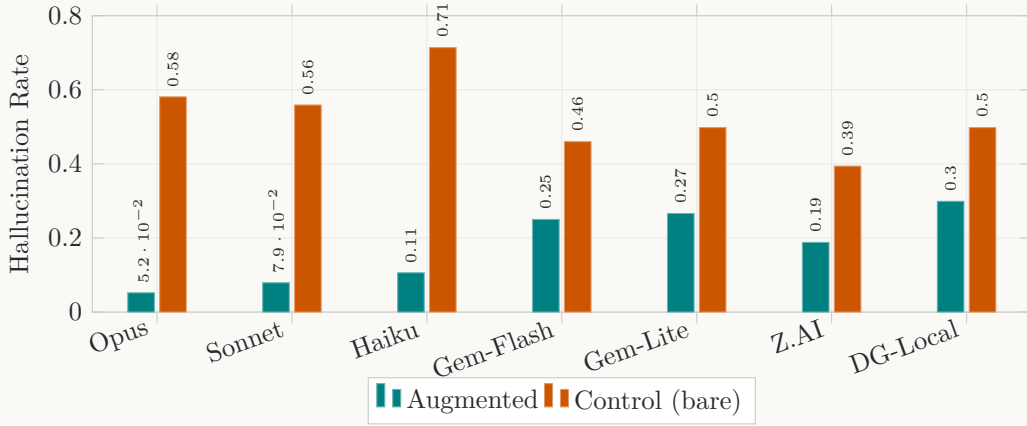


Figure 3: Ontology grounding reduces hallucination for every model. The largest reduction is Haiku ($0.714 \rightarrow 0.106$, -0.608). Even DiffusionGemma drops from 0.498 to 0.299.

7 Caveats on Lift Interpretation

The headline finding—that a local model with ontology grounding outperforms frontier models without it—requires careful interpretation. We detail the caveats below.

7.1 The comparison is grounded-local vs. bare-frontier, not like-for-like

DiffusionGemma+ontology (0.505) beats Opus bare (0.350), but this is not an apples-to-apples capability comparison. The local model receives structured KG triples as input context; the frontier models receive only the question. If frontier models were also grounded, they would score substantially higher: Opus AUG reaches 0.770, Sonnet AUG 0.755. The comparison demonstrates the *value of the ontology*, not the superiority of the local model’s reasoning.

7.2 Frontier models benefit far more from grounding

The ontology amplifies capability rather than substituting for it. Claude models gain +0.38–0.46 F1 from grounding; DiffusionGemma gains only +0.108. This asymmetry suggests that larger models are better at *exploiting* structured context—they extract more from the same triples. The ontology is a capability multiplier, and the multiplier scales with the base.

7.3 DiffusionGemma’s own augmentation lift is modest

At +0.108 F1, the local model’s lift is statistically meaningful but operationally small compared to the frontier lifts. The primary driver of DiffusionGemma’s competitive bare score (0.397) is surprisingly strong parametric knowledge—it already knows many of the domain concepts. The ontology’s contribution to the local model is more about *precision* (hallucination drops from 0.498 to 0.299) than about *recall*.

7.4 Performance varies sharply by question type

DiffusionGemma+ontology excels on existence queries (0.875, near ceiling) and subclass queries (0.500) but struggles on neighbour recall (0.323). The 256-token block decode window limits the number of distinct class names the model can enumerate per response. Frontier models with longer output windows (Opus neighbour AUG: 0.540) do not face this constraint. Headline F1 averages over these disparate regimes; no single number captures the full picture.

7.5 The condense pass is a one-off cost, not free

The ontology alias index was built by running DiffusionGemma over all 7,445 OWL classes (~6 hours on a single GPU). This is amortised over all future queries, but it means the system requires upfront investment. Re-indexing is needed when the KG changes materially.

7.6 Gold truth is the KG itself

The evaluation measures fidelity to the knowledge graph’s asserted ontology, not general-purpose question answering. Results do not transfer directly to open-domain QA benchmarks. A model that scores poorly here may still be excellent at tasks outside the KG’s domain.

7.7 Sample size and domain scope

16 questions $\times$ 3 reps $\times$ 7 models = 672 cells. While statistically powered for the observed effect sizes (Cohen’s $d > 1.0$ for Claude models), the question set is drawn from a single domain (DreamLab’s Web3/XR/AI ontology). Generalisation to other knowledge graphs and domains is plausible but untested.

8 Findings

1. **Ontology grounding is model-agnostic and universally beneficial.** All 7 models show positive F1 lift and reduced hallucination. The effect is strongest on niche, project-specific concepts where parametric knowledge is weakest.
2. **A local model + ontology can exceed frontier bare for domain-specific recall.** DiffusionGemma+ontology (0.505) beats every cloud model tested without grounding. This validates the thesis that structured retrieval compensates for raw parameter count [1, 6].
3. **Grounding amplifies capability—frontier models gain the most.** Opus AUG (0.770) is the highest score observed. The ontology is a force multiplier, not an equaliser (§7).
4. **Subclass recall is the largest relative win.** From 0.099 (CTL) to 0.671 (AUG)—a $6.8\times$ improvement. Models have almost no parametric knowledge of ontology hierarchies; the KG supplies it directly.
5. **Hallucination halves with grounding.** Overall $0.529 \rightarrow 0.177$, consistent with the KG-grounding literature [12, 13].
6. **Data sovereignty is achievable without sacrificing domain accuracy.** DiffusionGemma runs entirely on-premise with zero API cost. For regulated or privacy-sensitive deployments, this is the critical finding [19].

9 Threats to Validity

- Gold is the KG’s asserted graph; KG errors propagate to gold (mitigated by curating concepts with rich, sensible neighbourhoods).
- The grader’s token-set matching can over- or under-credit near-synonyms; thresholds were smoke-validated for non-saturation before the full run.
- The augmented arm uses the skill CLI, not the MCP tool directly; both share the same `createDefaultRetrieval` brain, so results transfer.
- DiffusionGemma’s 256-token block decode constrains neighbour enumeration; autoregressive models do not face this limit. A fairer neighbour comparison would cap output tokens.
- Gemini and Z.AI models were accessed via their respective commercial APIs; rate limits and non-determinism at temperature=0 may introduce variance not present in locally-hosted models.

References

- [1] Patrick Lewis et al. “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks”. In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2020. arXiv: 2005.11401 [cs.CL]. URL: <https://arxiv.org/abs/2005.11401>.
- [2] Darren Edge et al. “From Local to Global: A Graph RAG Approach to Query-Focused Summarization”. In: *arXiv preprint arXiv:2404.16130* (2024). arXiv: 2404.16130 [cs.CL]. URL: <https://arxiv.org/abs/2404.16130>.
- [3] Zirui Guo et al. “LightRAG: Simple and Fast Retrieval-Augmented Generation”. In: *arXiv preprint arXiv:2410.05779* (2024). arXiv: 2410.05779 [cs.IR]. URL: <https://arxiv.org/abs/2410.05779>.
- [4] Yuntong Hu et al. “GRAG: Graph Retrieval-Augmented Generation”. In: *arXiv preprint arXiv:2405.16506* (2024). arXiv: 2405.16506 [cs.CL]. URL: <https://arxiv.org/abs/2405.16506>.
- [5] Microsoft Research. *BenchmarkQED: Automated Benchmarking of RAG Systems*. Microsoft Research Blog. 2025. URL: <https://www.microsoft.com/en-us/research/blog/benchmarkqed-automated-benchmarking-of-rag-systems/>.
- [6] Zihan Liu et al. “ChatQA: Surpassing GPT-4 on Conversational QA and RAG”. In: *arXiv preprint arXiv:2401.10225* (2024). arXiv: 2401.10225 [cs.CL]. URL: <https://arxiv.org/abs/2401.10225>.
- [7] Zhuo Li et al. “Improving Retrieval Augmented Open-Domain Question-Answering with Vectorized Contexts”. In: *arXiv preprint arXiv:2404.02022* (2024). arXiv: 2404.02022 [cs.CL]. URL: <https://arxiv.org/abs/2404.02022>.
- [8] Marah Abdin et al. “Phi-3 Technical Report: A Highly Capable Language Model Locally on Your Phone”. In: *arXiv preprint arXiv:2404.14219* (2024). arXiv: 2404.14219 [cs.CL]. URL: <https://arxiv.org/abs/2404.14219>.
- [9] An Yang et al. “Qwen2 Technical Report”. In: *arXiv preprint arXiv:2407.10671* (2024). arXiv: 2407.10671 [cs.CL]. URL: <https://arxiv.org/abs/2407.10671>.
- [10] Gemma Team, Google DeepMind. “Gemma 3 Technical Report”. In: *arXiv preprint arXiv:2503.19786* (2025). arXiv: 2503.19786 [cs.CL]. URL: <https://arxiv.org/abs/2503.19786>.
- [11] Abhimanyu Dubey et al. “The Llama 3 Herd of Models”. In: *arXiv preprint arXiv:2407.21783* (2024). arXiv: 2407.21783 [cs.AI]. URL: <https://arxiv.org/abs/2407.21783>.

- [12] Garima Agrawal et al. “Can Knowledge Graphs Reduce Hallucinations in LLMs? A Survey”. In: *arXiv preprint arXiv:2311.07914* (2023). arXiv: 2311.07914 [cs.CL]. URL: <https://arxiv.org/abs/2311.07914>.
- [13] Hannah Sansford et al. “Can Knowledge Graphs Make Large Language Models More Trustworthy? An Empirical Study over Open-Ended Question Answering”. In: *arXiv preprint arXiv:2410.08085* (2024). arXiv: 2410.08085 [cs.CL]. URL: <https://arxiv.org/abs/2410.08085>.
- [14] Alon Jacovi et al. “The FACTS Grounding Leaderboard: Benchmarking LLMs’ Ability to Ground Responses to Long-Form Input”. In: *arXiv preprint arXiv:2501.03200* (2025). arXiv: 2501.03200 [cs.CL]. URL: <https://arxiv.org/abs/2501.03200>.
- [15] Shen Nie et al. “Large Language Diffusion Models”. In: *arXiv preprint arXiv:2502.09992* (2025). arXiv: 2502.09992 [cs.CL]. URL: <https://arxiv.org/abs/2502.09992>.
- [16] Inception Labs et al. “Mercury: Ultra-Fast Language Models Based on Diffusion”. In: *arXiv preprint arXiv:2506.17298* (2025). arXiv: 2506.17298 [cs.CL]. URL: <https://arxiv.org/abs/2506.17298>.
- [17] Google DeepMind. *Gemini Diffusion*. Google DeepMind model page. 2025. URL: <https://deepmind.google/models/gemini-diffusion/>.
- [18] Introl. *Inference Unit Economics: The True Cost Per Million Tokens*. Introl Blog. 2026. URL: <https://introl.com/blog/inference-unit-economics-true-cost-per-million-tokens-guide>.
- [19] Advisori. *GDPR-Compliant AI: On-Premise and EU LLMs vs ChatGPT*. Advisori Blog. 2026. URL: <https://www.advisori.de/en/blog/gdpr-compliant-ai-on-premise-llm-enterprise-guide>.